

(1001CJA102120106)

Test Pattern

**CLASSROOM CONTACT PROGRAMME**

(Academic Session : 2020 - 2021)

JEE(Advanced)
REVIEW TEST
26-06-2021
JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : ALL STAR BATCHES)**ANSWER KEY****PAPER-1****PART-1 : PHYSICS**

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C,D	D	A,B,C	A,B,C,D	A,C	B,D	A,B	B	D	A,C
SECTION-II	Q.	1	2	3	4	5	6	7	8		
	A.	720.00	262.50	4.00	240.50 to 240.70	2.00	111.11	27.00	336.00		
SECTION-IV	Q.1	A	B	C	D	Q.2	A	B	C	D	
		S,T	P	T	Q,R,T		P,Q,R,S	P	QRT	S	

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,B,D	A	A,C	A,B	B,C,D	A,C,D	B,C,D	A,D	A,B,D	A,C,D
SECTION-II	Q.	1	2	3	4	5	6	7	8		
	A.	0.02	8.10	33.30	54.00	5.00	2.00	+722.00	+11.00		
SECTION-IV	Q.1	A	B	C	D	Q.2	A	B	C	D	
		R	P,R	P,S	P,Q,R,S,T		S,T	Q,R,S	Q	P,Q,S	

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C,D	A	C	C	C	A,B,C	A,C,D	A,B	C	A,B
SECTION-II	Q.	1	2	3	4	5	6	7	8		
	A.	12.00	4.00	50.00	45.00	9.00	1.50	49.00	3.00		
SECTION-IV	Q.1	A	B	C	D	Q.2	A	B	C	D	
		Q	R	P	T		R,S,T	Q	P	P,R,S,T	

(1001CJA102120107)

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**CLASSROOM CONTACT PROGRAMME**

(Academic Session : 2020 - 2021)

JEE(Advanced)
REVIEW TEST
26-06-2021
JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : ALL STAR BATCHES)**ANSWER KEY****PAPER-2****PART-1 : PHYSICS**

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B,C	B,C,D	A,B,D	A,B,C	A,B,C,D	A,B,C	A,B,C	A,B,C	B	A,B,C
	Q.	11	12	13	14	15	16				
	A.	A	B	C	D	B	D				
SECTION-II	Q.	1	2	3	4						
	A.	5.00	3.00	11.00	3.00						

PART-2 : CHEMISTRY

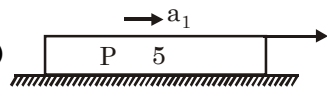
PART - I (Common)											
SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B,C,D	C	A,C,D	B,D	B,C	A,B,C	A,B,C	A,B,D	B,C	A,B,C,D
	Q.	11	12	13	14	15	16				
	A.	D	D	A	B	B	D				
SECTION-II	Q.	1	2	3	4						
	A.	3.00	4.00	3.00	+24.00						

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,B,D	A,B,C,D	A,B,D	B	A,B,D	A,C,D	A,B,D	A,B,C,D	A,C,D	A,B,C,D
	Q.	11	12	13	14	15	16				
	A.	C	D	D	C	C	C				
SECTION-II	Q.	1	2	3		4					
	A.	0.20	1.50	1.33 or 1.34		1.33 or 1.34					

PAPER-1**PART-1 : PHYSICS****SOLUTION****SECTION-I**

1. Ans. (C,D)

Sol. (C) 

$$10 = 5a,$$

$$\therefore a_1 = 2$$

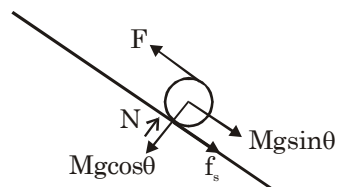
$$(D) 15 = (2 + 5)a_c$$

2. Ans. (D)

Sol. $mgr = \frac{1}{2}\mu v_r^2$

$$\therefore v_r = \sqrt{\frac{m+M}{M} 2gr}$$

3. Ans. (A,B,C)

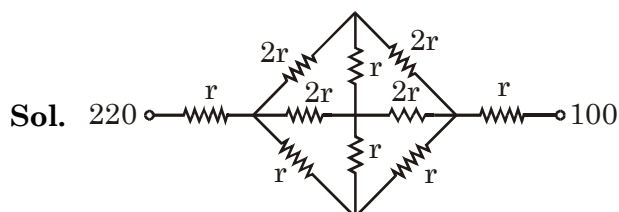
Sol. 

$$(A) F2R = Mgsin\theta R$$

$$(B) F = f_s + Mgsin\theta$$

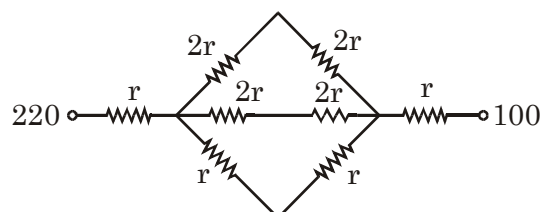
$$(C) f_s = \mu N$$

4. Ans. (A,B,C,D)



$$r = \frac{L}{KA}$$

From symmetry.

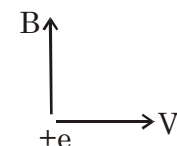


Now solve

5. Ans. (A, C)

Sol. By putting the value of x in electric field function, find the net force on dipole. If $\vec{P} \parallel \vec{F}$, then it will be stable equilibrium.

6. Ans. (B, D)

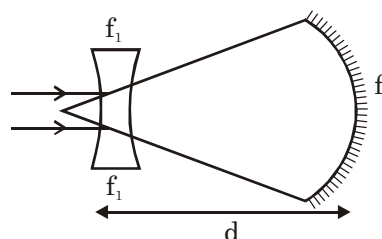
Sol. 

$\therefore E$ must be in $-z$ direction only
& $E = VB$

7. Ans. (A,B)

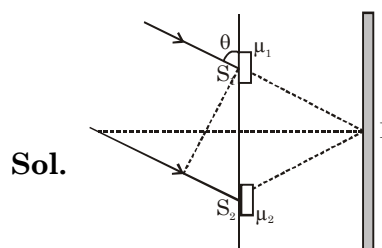
8. Ans. (B)

Sol. $(d + f_1) = 2f_2$; $d = 2|f_2| - |f_1|$



9. Ans. (D)

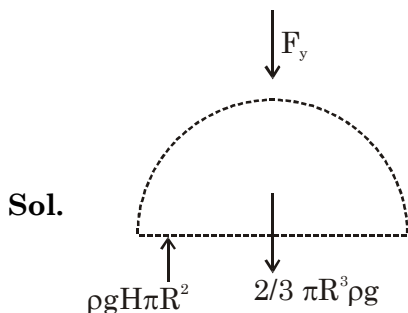
10. Ans. (A, C)



$$\Delta p = d \cos \theta + (S_2 P - t) + \mu_2 t - [(S_1 P - t) + \mu_1 t]$$

SECTION-II

1. Ans. 720.00



$$\begin{aligned} \therefore F_y &= \rho g H \pi R^2 - \frac{2}{3} \pi R^3 \rho g \\ &= \pi (0.3)^2 \times 1 \times 10^3 \times 10 - (0.3)^3 \times 10^3 \times 10 \\ &= 720\pi \end{aligned}$$

2. Ans. 262.50

Sol. $f = \frac{P}{2L} \sqrt{\frac{mg}{\mu}} \quad \dots(i)$

$$P\sqrt{9} = (P-1)\sqrt{25} \quad \dots(ii)$$

$$\therefore P = 2.5$$

$$\therefore P\sqrt{9} = (P-2)\sqrt{25}$$

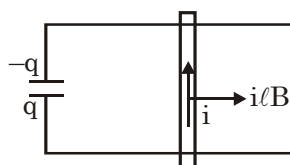
$$\therefore P = 5$$

$$\therefore f = \frac{5}{2 \times 2} \sqrt{\frac{90}{0.002}}$$

3. Ans. 4.00

Sol. $\ell B \int \frac{dq}{dt} = m \int \frac{dv}{dt}$

In steady state, $\ell B(Q - q) = mv$



$$CB\ell v = q$$

$$\ell BQ - C\ell^2 B^2 v = mv$$

$$v = \frac{B\ell Q}{m + CB^2\ell^2}$$

$$K = \frac{1}{2}mv^2 = \frac{1}{2}m \left(\frac{B\ell Q}{m + CB^2\ell^2} \right)^2$$

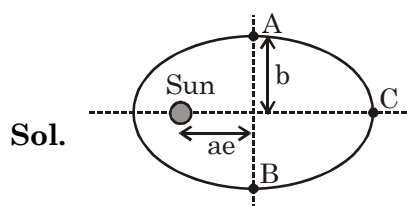
$$\frac{dk}{dm} = 0 = \frac{1}{2} \left(\frac{B\ell Q}{m + CB^2\ell^2} \right)^2$$

$$= -\frac{2}{2} \frac{m(B\ell Q)^2}{(m + CB^2\ell^2)^3} \times 1 \frac{1}{2} - \frac{m}{m + CB^2\ell^2} = 0$$

$$CB^2\ell^2 = m$$

$$\Rightarrow m = 4 \times 10^{-6} \text{ kg} = 4\text{mg}$$

4. Ans. 240.50 to 240.70



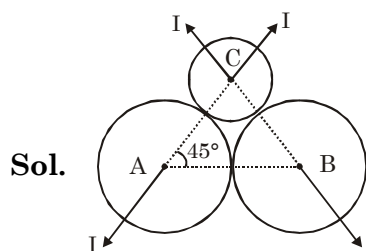
$$\text{Area covered} = \frac{1}{2}ae \times b + \frac{\pi ab}{2}$$

$$= \frac{ab}{2}(e + \pi)$$

$$\pi ab \rightarrow T$$

$$\text{then } \frac{ab}{2} \left(\frac{1}{2} + \pi \right) \rightarrow T'$$

5. Ans. 2.00



$$I \cos \theta = m_A v_A = m_B v_B$$

$$\Rightarrow I = \frac{m_A v_A}{\cos \theta}$$

$$= \frac{2 \times 3}{\cos 45^\circ}$$

$$= 6\sqrt{2}$$

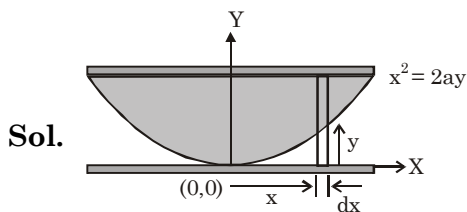
For ball C

$$m_C v'_C = m_C v_C - 2I \sin \theta$$

$$\Rightarrow v'_C = v_C - \frac{2I}{m_C} \sin 45^\circ$$

$$= 10 - \frac{2 \times 6\sqrt{2}}{1} \times \frac{1}{\sqrt{2}} = -2 \text{ m/s}$$

6. Ans. 111.11



$$dc = \frac{\epsilon_0 dA}{\frac{a}{8} - y \left(1 - \frac{1}{k}\right)}$$

$$= \frac{\epsilon_0 a dx}{\frac{a}{8} - \frac{x^2}{2a} \left(1 - \frac{1}{2}\right)} = \frac{4\epsilon_0 a dx}{\frac{a}{2} - \frac{x^2}{a}}$$

$$\therefore C = 2 \int_0^{a/2} \frac{4\epsilon_0 a dx}{\frac{a}{2} - \frac{x^2}{a}}$$

$$= 8\epsilon_0 a (1.2465)$$

7. Ans. 27.00

Sol. $\vec{\tau} = \vec{P} \times \vec{E}$

$$\Rightarrow I\alpha = \left[\left(\frac{\lambda \ell}{2} \right) \frac{\ell}{2} \times \frac{\sigma}{2\epsilon_0} \right]$$

$$\alpha = \frac{\frac{\lambda \ell^2}{4} \times \frac{\sigma}{2\epsilon_0}}{\frac{m \ell^2}{12}}$$

8. Ans. 336.00

Sol. $\pi(4)^2 10 = \pi(2)^2 \Delta \ell$ (i)

$$f = \frac{1}{4\ell_1} C$$
(ii)

$$f = \frac{3}{4\ell_2} C$$
(iii)

$$\Delta \ell = \ell_2 - \ell_1$$
(iv)

SECTION-IV

1. Ans. (A)-S,T; (B)-P; (C)-T; (D)-Q,R,T

Sol. For highly conducting container, system and surrounding temperature will be same.

2. Ans. (A)-P,Q,R,S; (B)-P; (C)-Q,R,T; (D)-S

Sol. (A) C_1 and L_1 are in series, hence V_{L_1} and

V_{C_1} will be out of phase.

Also if $R_1 = R_2$ and both are in series,

$$\therefore V_{R_1} = V_{R_2}$$

(B) C_1 and L_1 are in series, hence V_{L_1} and

V_{C_1} will be out of phase

(C) If $R_1 = R_2$, then $V_{R_1} = V_{R_2}$

(D) C_2 and L_1 are in series, hence V_{C_2} and

V_{L_1} will be out of phase.

SECTION-I

1. Ans.(A,B,D)

$$\frac{\frac{m'_{\text{He}}}{4}}{\frac{m'_{\text{CH}_4}}{16}} = \frac{m \times 16}{4 \times m} \times \sqrt{\frac{16}{4}} : \frac{m'_{\text{He}}}{m'_{\text{CH}_4}} = \frac{2}{1}$$

2. Ans.(A)

$$\frac{\lambda^2}{4\pi} = \frac{1}{2\pi}$$

$$\lambda = \sqrt{2} = \sqrt{\frac{150}{V}}$$

3. Ans.(A,C)

4. Ans.(A,B)

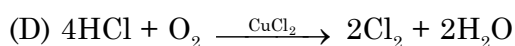
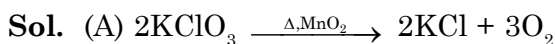
5. Ans.(B,C,D)

6. Ans.(A,C,D)

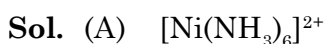
7. Ans.(B,C,D)

Sol. Lone pair of nitrogen is not delocalised in $(\text{C}_2\text{H}_5)_3\text{N}$, because back bond is not present. But in other species back bond is present.

8. Ans.(A,D)



9. Ans.(A,B,D)



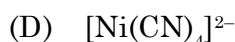
$$\mu = \sqrt{8} \text{ B.M.}$$



$$\mu = \sqrt{8} \text{ B.M.}$$



$$\mu = \sqrt{8} \text{ B.M.}$$



$$\mu = 0$$

10. Ans.(A,C,D)

$$\log k = \frac{-E_a}{2.303R} \left(\frac{1}{T} \right) + \log A$$

SECTION-II

1. Ans.(0.02)

No. of molecules in 'V' litre

$$= \frac{5.12 \times V}{256} \times 6 \times 10^{23} = 1.2 \times 10^{22} V$$

$$\text{Area covered} = (1.2 \times 10^{22} V \times 0.2 \times 10^{-14}) \text{ cm}^2$$

$$= 2.4 \times 10^7 V \text{ cm}^2$$

$$2.4 \times 10^7 V = 480$$

$$V = 2 \times 10^{-5} \text{ litre}$$

$$V = 0.02 \text{ ml}$$

2. Ans.(8.10)

$$\Lambda_m^\infty(\text{AgCl}) = 62 + 78 = 140$$

$$140 = \frac{1.26 \times 10^{-6} \times 1000}{s}$$

$$s = 9 \times 10^{-6}$$

$$K_{sp} = s^2 = 81 \times 10^{-12} = 8.1 \times 10^{-11}$$

3. Ans.(33.30)

$$\frac{W}{180} = (1 + 0.85) \times 0.1$$

$$W = 33.3$$

4. Ans.(54.00)

5. Ans.(5.00)

6. Ans.(2.00)

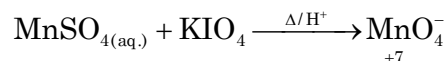
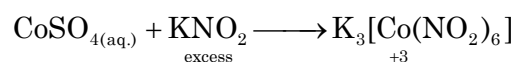
7. Ans.(+722.00)

$$\begin{aligned} \text{Sol. X} &= \text{Molar mass of } [\text{Co}(\text{NH}_3)_5(\text{H}_2\text{O})]\text{Br}_3 \\ &= 59 + (17 \times 5) + 18 + (80 \times 3) \\ &= 402 \end{aligned}$$

$$\begin{aligned} \text{Y} &= \text{Molar mass of } [\text{Co}(\text{NH}_3)_5\text{Br}]\text{SO}_4 \\ &= 59 + (17 \times 5) + 80 + 32 + 64 \\ &= 320 \end{aligned}$$

$$X + Y = 402 + 320 = 722.00$$

8. Ans.(+11.00)



$$1 + 3 + 7 = 11.00$$

SECTION-IV

1. Ans. (A)→(R); (B)→(P,R); (C)→(P,S); (D)→(P,Q,R,S,T)

2. Ans.(A) - S, T ; (B) - Q, R, S ; (C) - Q ; (D) - P, Q, S

Sol. (A) $\text{NO}[\text{BF}_4]$

BF_4^- is tetrahedral and bond order of NO^+ is 3.0 , So NO^+ contain π -bond.

(B) $(\text{NH}_4)_2[\text{NiCl}_4]$

$[\text{NiCl}_4]^{2-}$ is sp^3 hybrid, paramagnetic having tetrahedral geometry.

NH_4^+ is also tetrahedral

(C) CsO_2

O_2^- is paramagnetic

(D) $[\text{Co}(\text{H}_2\text{O})_6][\text{Co}(\text{Cl})_4]$

$[\text{Co}(\text{H}_2\text{O})_6]^{2+}$ is sp^3d^2 hybrid, octahedral and paramagnetic

$[\text{Co}(\text{Cl})_4]^{2-}$ is sp^3 hybrid, tetrahedral and paramagnetic

PART-3 : MATHEMATICS

SOLUTION

SECTION-I

1. **Ans. (C,D)**

Sol. $f(f(n)) = n$

$$\text{put } n = 0 \Rightarrow f(f(0)) = 0$$

$$f(1) = 0$$

$$\text{since } f(f(n+2)+2) = n \quad \dots(1)$$

$$\text{put } n = -1$$

$$f(2) = -1$$

$$\text{put } n = -2 \text{ in equation (1)} \Rightarrow f(3) = -2$$

2. **Ans. (A)**

Sol. $\lim_{x \rightarrow \infty} \left(\frac{x+1}{x+2} \right)^x$

$$\lim_{x \rightarrow \infty} \left(1 - \frac{1}{x+2} \right)^x$$

$$e^{\lim_{x \rightarrow \infty} \left(\frac{-x}{x+2} \right)} = \frac{1}{e}$$

3. **Ans. (C)**

Sol. $\therefore$ at $k = n\pi$, $f(x) = [x](\sin n\pi x)^p$ is always continuous in its domain for $p \in \mathbb{R} - \{0\}$

4. **Ans. (C)**

Sol. $f(x+1) = x + f(x)$

$$e^{f(x+1)} = e^{x+f(x)} = e^x \cdot e^{f(x)}$$

$$g(x+1) = e^x g(x)$$

$$\ln g(x+1) = x + \ln g(x)$$

$$\frac{g'(x+1)}{g(x+1)} = 1 + \frac{g'(x)}{g(x)}$$

$$\therefore \frac{g'(x+1)}{g(x+1)} - \frac{g'(x)}{g(x)} = 1$$

$$\text{Put } x = \frac{1}{2}, \frac{g'\left(\frac{1}{2}+1\right)}{g\left(\frac{1}{2}+1\right)} - \frac{g'\left(\frac{1}{2}\right)}{g\left(\frac{1}{2}\right)} = 1$$

$$\text{Put } x = \frac{3}{2}, \frac{g'\left(\frac{1}{2}+2\right)}{g\left(\frac{1}{2}+2\right)} - \frac{g'\left(\frac{3}{2}\right)}{g\left(\frac{3}{2}\right)} = 1$$

$\vdots$
 $\vdots$

$$\text{Put } x = n - \frac{1}{2}, \frac{g'\left(n + \frac{1}{2}\right)}{g\left(n + \frac{1}{2}\right)} - \frac{g'\left(n - \frac{1}{2}\right)}{g\left(n - \frac{1}{2}\right)} = 1$$

add

$$\frac{g'\left(n + \frac{1}{2}\right)}{g\left(n + \frac{1}{2}\right)} - \frac{g'\left(\frac{1}{2}\right)}{g\left(\frac{1}{2}\right)} = n$$

5. **Ans. (C)**

Sol. $\phi(b) = 0$, since Rolle's theorem is applicable

$$\therefore \phi(a) = 0$$

$$\Rightarrow A = \frac{(f(b) - f(a)) - (b - a)f'(a)}{(b - a)^2} \quad \dots(1)$$

$$\phi'(c) = 0$$

$$\Rightarrow A = \frac{1}{2} f''(c) \quad \dots(2)$$

from (1) and (2)

$$f(b) = f(a) + (b - a)f'(a) + \frac{1}{2}(b - a)^2 f''(c)$$

$$\lambda = \frac{1}{2}$$

6. **Ans. (A,B,C)**

Sol. Let $g(x) = \frac{f(x)}{x}$

$$g'(x) = \frac{xf'(x) - f(x)}{x^2}$$

$$= \underbrace{\frac{f'(x)}{x^2}}_{\text{positive}} \underbrace{\left(x - \frac{f(x)}{f'(x)} \right)}_{h(x)}$$

$$\therefore h(x) = x - \frac{f(x)}{f'(x)}$$

$$h'(x) = 1 - \frac{(f'(x)f'(x) - f(x)f''(x))}{(f'(x))^2}$$

$$h'(x) = \frac{f(x)f''(x)}{(f'(x))^2} > 0$$

$h(x)$ is increasing

$$\therefore h(x) > h(0) \quad \forall x \in (0, \infty)$$

$$\therefore x - \frac{f(x)}{f'(x)} > 0 - \frac{f(0)}{f'(0)}$$

$$\therefore x - \frac{f(x)}{f'(x)} > 0$$

$$h(x) > 0 \quad \forall x \in (0, \infty)$$

$$\therefore g'(x) > 0 \quad \forall x \in (0, \infty)$$

$\therefore g(x)$ is increasing

$$\therefore g(\alpha) < g(\beta) \Rightarrow \frac{f(\alpha)}{\alpha} < \frac{f(\beta)}{\beta}$$

$$\therefore g(\gamma) < g(\gamma + \beta - \alpha) \Rightarrow \frac{f(\gamma)}{\gamma} < \frac{f(\gamma + \beta - \alpha)}{\gamma + \beta - \alpha}$$

7. Ans. (A,C,D)

$$\text{Sol. } f(x) = \int_0^x (5 \ln(1+t^2) - 10t \tan^{-1} t + 16 \sin t) dt$$

$$f'(x) = 5 \ln(1+x^2) - 10x \tan^{-1} x + 16 \sin x$$

$$f''(x) = \frac{5}{1+x^2} (2x) - 10 \left(x \cdot \frac{1}{1+x^2} + \tan^{-1} x \right) + 16 \cos x$$

$$f''(x) = -10 \tan^{-1} x + 16 \cos x$$

$$f'''(x) = -16 \sin x - \frac{10}{1+x^2}$$

$$\Rightarrow - \left(16 \sin x + \frac{10}{1+x^2} \right) < 0 \quad \forall x \in (0, 1)$$

$\therefore f''(x)$ is decreasing

$$f''(x) > f''(1)$$

$$f''(x) > -10 \left(\frac{\pi}{4} \right) + 16 \cos 1$$

$$> 2 \left(8 \cos 1 - \frac{5\pi}{4} \right)$$

$$> 2 \left(8 \cos \frac{\pi}{3} - \frac{5\pi}{4} \right)$$

$$> 2 \left(4 - \frac{5\pi}{4} \right) > 0$$

$$\therefore f''(x) > 0$$

$\therefore f'(x)$ is increasing

$$f'(x) > f'(0)$$

$$f'(x) > 0$$

$\therefore f(x)$ is increasing $\forall x \in (0, 1)$

$$\therefore f(x) > f(0)$$

$$f(x) > 0$$

$\therefore f(x)$ is also positive and $\int_0^x f(t) dt$ is increasing

8. Ans. (A,B)

$$\text{Sol. } xy + 4x - 3y - 12 = (x-3)(y+4) = 0$$

$$xy - 3x + 4y - 12 = (x+4)(y-3) = 0$$

$\therefore$ vertices are A(-4, -4), B(-4, 3), C(3, 3), D(3, -4)

$\therefore$ diagonal AC is $x = y$ and diagonal BD is $x + y + 1 = 0$

9. Ans. (C)

$$\text{Sol. } \omega_1 : (x-a)^2 + (y-b)^2 = a^2 + b^2 \quad \dots(1)$$

$$\omega_2 : (x-b)^2 + (y-a)^2 = a^2 + b^2 \quad \dots(2)$$

Since origin is satisfied by both circles

Eq. (1) - (2)

$$(2x-a-b)(b-a) + (2y-a-b)(a-b) = 0$$

$$y = x$$

$\therefore$ Radical axis is $y = x$, hence circles cut at another point which lies on $y = x$.

10. Ans. (A,B)

Sol. Let P($a \cos \theta$, $b \sin \theta$), Q($-a \sin \theta$, $b \cos \theta$)

equation of tangent at P :

$$\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1 \quad \dots(1)$$

equation of tangent at Q :

$$\frac{-x \sin \theta}{a} + \frac{y \sin \theta}{b} = 1 \quad \dots(2)$$

$$(1)^2 - (2)^2 :$$

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 2$$

midpoint of P and Q is

$$R \left(\frac{a(\cos \theta - \sin \theta)}{2}, \frac{b(\sin \theta + \cos \theta)}{2} \right)$$

$$\therefore x = \frac{a}{2} (\cos \theta - \sin \theta)$$

$$y = \frac{b}{2} (\sin \theta + \cos \theta)$$

$$\Rightarrow \frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{1}{2}$$

SECTION-II

1. Ans. 12.00

Sol. Let α & β are common roots

$$\therefore \alpha^3 + 5\alpha^2 + p\alpha + q = 0 \quad \dots(1)$$

$$\alpha^3 + 7\alpha^2 + p\alpha + r = 0 \quad \dots(2)$$

$$2\alpha^2 + r - q = 0 \text{ has roots } \alpha \text{ \& } \beta$$

$$\Rightarrow \alpha + \beta = 0$$

$$\text{since } \alpha + \beta + \gamma_1 = -5$$

$$\alpha + \beta + \gamma_2 = -7$$

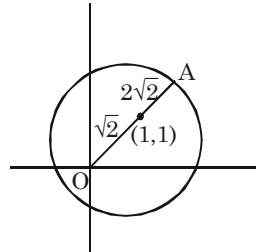
$$\Rightarrow \gamma_1 + \gamma_2 = -12$$

2. **Ans. 4.00**

Sol. $|z - (1 + i)| = 2\sqrt{2}$

$$|z|_{\max} = OA = 3\sqrt{2}$$

$$[|z|] = 4$$



3. **Ans. 50.00**

Sol. Let equation of the plane is

$$(x + 2y + z) + \lambda(2x - y - z - 1) = 0$$

since it passes through (1, 2, 1)

$$\Rightarrow 6 + (-2)\lambda = 0 \Rightarrow \lambda = 3$$

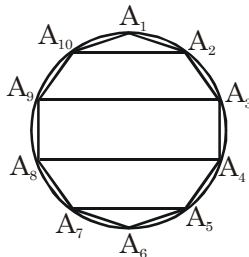
$\therefore$ equation of plane P is : $7x - y - 2z - 3 = 0$

$$d = \left| \frac{14 - 1 - 3}{\sqrt{49 + 1 + 4}} \right| = \left| \frac{10}{\sqrt{54}} \right|$$

$$d^2 = \frac{50}{27}$$

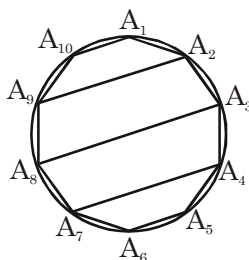
4. **Ans. 45.00**

Sol.



$${}^5C_1 \times {}^4C_2 = 30$$

No. of ways to select a vertex and No. of ways to select a pair



$$= 3C_2 \times 5 = 15$$

$\therefore$ Ans. = 45

5. **Ans. 9.00**

Sol. $P_n = {}^nC_0 \cdot {}^nC_1 \cdot {}^nC_2 \dots \dots {}^nC_n$

$$P_{n+1} = {}^{n+1}C_0 \cdot {}^{n+1}C_1 \cdot {}^{n+1}C_2 \dots \dots {}^{n+1}C_{n+1}$$

$$\begin{aligned} \therefore \frac{P_{n+1}}{P_n} &= \frac{{}^{n+1}C_0}{{}^nC_0} \cdot \frac{{}^{n+1}C_1}{{}^nC_1} \cdot \frac{{}^{n+1}C_2}{{}^nC_2} \dots \dots \frac{{}^{n+1}C_n}{{}^nC_n} \cdot {}^{n+1}C_{n+1} \\ &= (1) \left(\frac{n+1}{n} \right) \left(\frac{n+1}{n-1} \right) \left(\frac{n+1}{n-2} \right) \dots \dots \frac{n+1}{1} \\ &= \frac{(n+1)^n}{\underline{n}} = \frac{10^9}{\underline{9}} \end{aligned}$$

$$\Rightarrow n = 9$$

6. **Ans. 1.50**

Sol. $A^2 = A$

$$(I - 0.4A)^{-1} = (I - \alpha A)$$

$$I = (I - 0.4A)(I - \alpha A)$$

$$I = I - \alpha A - 0.4A + 0.4\alpha A^2$$

$$0.4\alpha A - 0.4A - \alpha A = 0$$

$$A(0.4\alpha - 0.4 - \alpha) = 0$$

$$\therefore 0.4\alpha - \alpha = 0.4$$

$$-0.6\alpha = 0.4$$

$$\alpha = \frac{-2}{3}$$

$$\Rightarrow \frac{1}{\alpha} = \frac{-3}{2} \Rightarrow \left| \frac{1}{\alpha} \right| = 1.50$$

7. **Ans. 49.00**

Sol. $ax^4 + ax^2 + bx^3 + bx + x^2 + 1$

$$= (x^2 + 1)(ax^2 + bx + 1)$$

$\therefore$ ' $x^2 + 1$ ' is positive for all real x

$$\text{For } ax^2 + bx + 1 > 0 \forall x \in \mathbb{R}$$

$$\therefore a > 0 \text{ \& } b^2 - 4a < 0$$

b No. of values a can take

1 9

2 8

3 7

4 5

5 3

$$\text{Probability} = \frac{32}{81}$$

8. Ans. 3.00

Sol. $\vec{a}, \vec{b}, |\vec{a}| = 2, |\vec{b}| = 3$

$$\vec{c} \times \vec{a} = \vec{b} \Rightarrow |\vec{c} \times \vec{a}| = |\vec{b}|$$

$$\Rightarrow |\vec{c}| |\vec{a}| \sin \theta = |\vec{b}|$$

$$|\vec{c}| = \frac{3}{2 \sin \theta}$$

$$2|\vec{c} - \vec{a}| = \sqrt{9 \operatorname{cosec}^2 \theta + 16 - 24 \cot \theta}$$

$$= \sqrt{9 \cot^2 \theta - 24 \cot \theta + 25}$$

$$= \sqrt{(3 \cot \theta - 4)^2 + 9}$$

$$\Rightarrow 2|\vec{c} - \vec{a}|_{\text{least}} = 3$$

SECTION-IV

1. Ans. (A)→(Q); (B)→(R); (C)→(P); (D)→(T)

Sol. $\cos A = \frac{b^2 + c^2 - a^2}{2bc}$

$$\Rightarrow c^2 - (2bc \cos A)c + b^2 - a^2 = 0$$

$$\therefore c_1 + c_2 = 2b \cos A, c_1 c_2 = b^2 - a^2$$

$$\Delta_1 = \frac{1}{2} b c_1 \sin A, \Delta_2 = \frac{1}{2} b c_2 \sin A$$

$$A.M. = \frac{\Delta_1 + \Delta_2}{2} = \frac{b}{4} (c_1 + c_2) \sin A$$

$$= \frac{1}{4} (2b^2 \cos A) \sin A = \frac{b^2}{4} \sin 2A$$

$$G.M. = \sqrt{\Delta_1 \Delta_2} = \sqrt{\frac{1}{4} b^2 (b^2 - a^2) \sin^2 A} = \frac{1}{2} b \sin A \sqrt{b^2 - a^2}$$

$$H.M. = \frac{2\Delta_1 \Delta_2}{\Delta_1 + \Delta_2} = \left(\frac{b^2 - a^2}{2} \right) \tan A$$

$$|\Delta_1 - \Delta_2| = \frac{1}{2} b \sin A \sqrt{4b^2 \cos^2 A - 4(b^2 - a^2)}$$

$$= b \sin A \sqrt{a^2 - b^2 \sin^2 A}$$

2. Ans. (A)→(R,S,T); (B)→(Q); (C)→(P); (D)→(P,R,S,T)

Sol. from (1) and (2) equation

$$x = z, y = 3 - 2z$$

putting this in last equation

$$(\lambda - 5)z = m - 9$$

If $\lambda \neq 5$

⇒ system of equation has unequal solution and consistent

when $\lambda = 5$ and $m \neq 9$

⇒ no solution

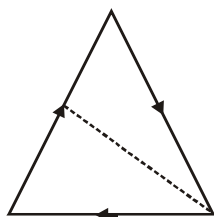
when $\lambda = 5$ and $m = 9$

⇒ system has infinite solution

PAPER-2**PART-1 : PHYSICS****SOLUTION****SECTION-I**

1. Ans. (B,C)

Sol.



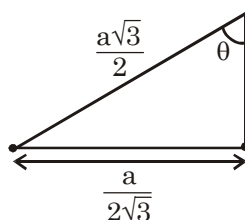
$$B = \frac{\mu_0 I}{4\pi d} (\sin \theta_1 + \sin \theta_2)$$

$$d = \frac{a}{2\sqrt{3}} \Rightarrow \theta_1 = \theta_2 = 60^\circ$$

$$B_{\text{net}} = 3B$$

at axis

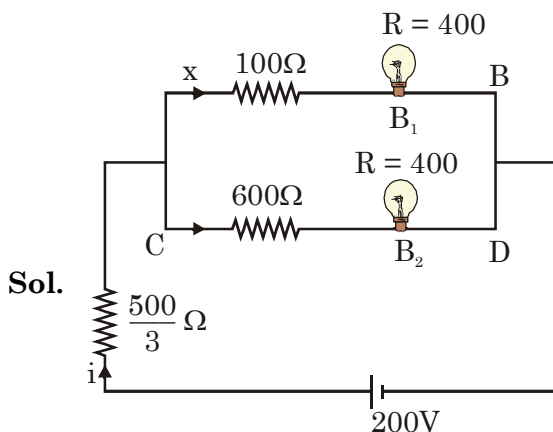
$$B = \frac{\mu_0 I}{4\pi \left(\frac{a\sqrt{3}}{2} \right)} [\sin(30) + \sin(30)]$$



$$\sin \theta = \frac{1}{3}$$

$$B_{\text{net}} = 3(B \sin \theta)$$

2. Ans. (B,C,D)



$$R_{\text{eq}} = 500$$

$$i = \frac{2}{5}$$

$$x = \frac{4}{15}$$

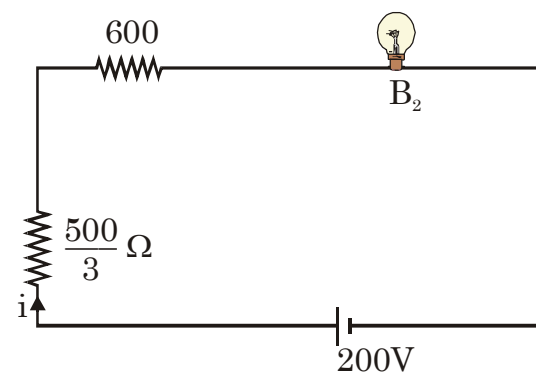
$$y = \frac{2}{15}$$

voltage across $B_1 = x \times 400$

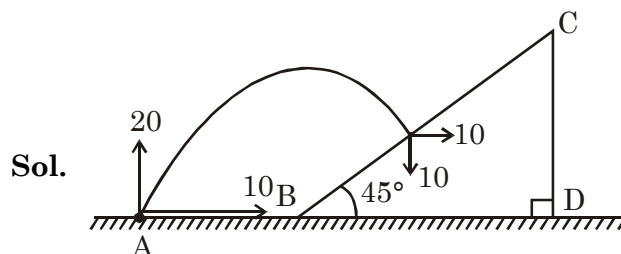
$$= \frac{1600}{15} > 100$$

 B_1 will fuse

New circuit



3. Ans. (A,B,D)



Sol.

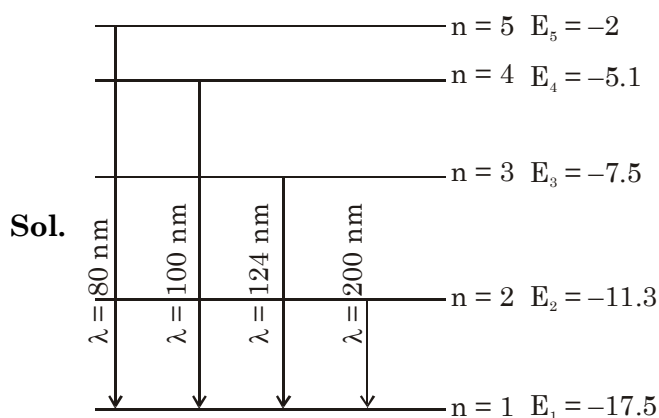
4. Ans. (A,B,C)

$$\text{Sol. } \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \text{ equation of ellipse}$$

$$ROC = \frac{\left(1 + \left(\frac{dy}{dx}\right)^2\right)^{3/2}}{d^2y/dx^2}$$

$$a_c = \frac{V^2}{ROC}$$

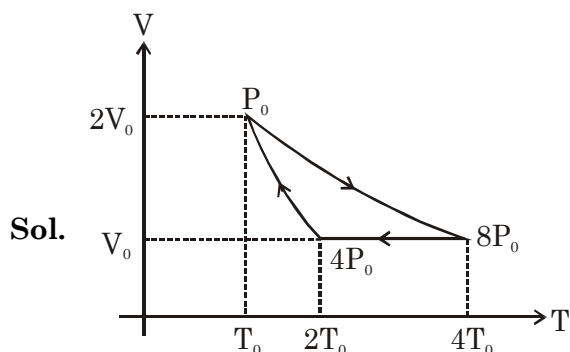
5. Ans. (A,B,C,D)



6. Ans. (A,B,C)

Sol. $\lambda_{ap} = \left(\frac{V \pm V_s}{V}\right) \lambda$

7. Ans. (A,B,C)



$$AB \Rightarrow TV^2 = \text{constant}$$

$$CA \Rightarrow TV = \text{constant}$$

A to B

$$PV^3 = \text{constant}$$

$$CP = \frac{3}{2}R + \frac{R}{1-3}$$

$$\Delta Q = nC_{\text{Process}} \Delta T = 3R(T_B - T_A)$$

$$= 8P_0 V_0 - 2V_0 P_0 = 6P_0 V_0$$

C to A

$$PV^2 = \text{constant}$$

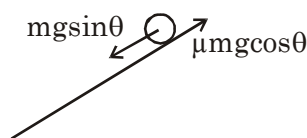
$$W = \frac{nR\Delta T}{1-2} = \frac{nR(T_A - T_C)}{-1}$$

$$= 4P_0 V_0 - 2P_0 V_0 = 2P_0 V_0$$

8. Ans. (A,B,C)

Sol. Pure rolling or no friction $\Rightarrow$ no heat is dissipated.

for pure rolling, $\mu \geq \frac{\tan \theta}{1 + \frac{mR^2}{I}} = \frac{1}{4}$



$$a = g \sin \theta - \mu g \cos \theta$$

$$\alpha = \frac{\mu mg \cos \theta R}{\frac{1}{2} m R^2} = \frac{2\mu g \cos \theta}{R}$$

heat dissipated is max.
when KE at bottom is min.

$$k_{\text{bottom}} = \frac{1}{2} m v^2 + \frac{1}{2} I \omega^2$$

$$= \frac{1}{2} \times m a^2 t^2 + \frac{1}{2} \times \frac{1}{2} m R^2$$

$$= \frac{a t^2}{2} \left[m a + \frac{1}{2} \frac{m R^2 \alpha^2}{a} \right]$$

$$= S[mg \sin \theta - \mu mg \cos \theta] + 2 \times \frac{\mu^2 g^2 \cos^2 \theta}{g \sin \theta - \mu g \cos \theta}$$

$$k = mgh \left[1 - \mu \times \frac{4}{3} + \frac{\mu^2 \times \frac{32}{9}}{1 - \frac{4\mu}{3}} \right]$$

$$\frac{dk}{d\mu} = 0 \Rightarrow \mu = \frac{3}{4} \left(1 - \sqrt{\frac{2}{3}} \right)$$

9. Ans. (B)

10. Ans. (A,B,C)

Sol. $L = 50\text{m}$, $d = 1 + 0.25 = 1.25$

$$\Delta d = 10^{-2} \text{ mm}$$

$$\Delta \ell = 10^{-3} \text{ cm}$$

$$\Delta L = 10^{-2} \text{ cm}$$

$$Y = \frac{4FL}{\pi D^2 \times \ell}$$

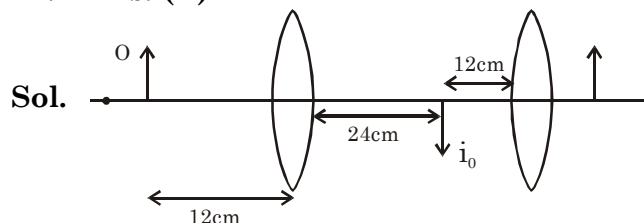
$$\frac{dY}{Y} = -2 \frac{d(D)}{D} + \frac{dL}{L} + \frac{d\ell}{\ell}$$

$$\frac{dY}{Y} \times 100 = \frac{2 \times 10^{-2}}{1.25} + \frac{10^{-2}}{50} + \frac{10^{-3}}{0.125} \times 100$$

$$= \frac{2}{1.25} + \frac{1}{50} + \frac{0.1}{0.125} = 1.6 + 0.02 + 0.8 = 2.4$$

11. Ans. (A)

12. Ans. (B)



$$\frac{1}{V} - \frac{1}{-12} = \frac{1}{8}$$

$$\frac{1}{V} = \frac{1}{8} - \frac{1}{12}$$

$$\frac{1}{V} = \frac{6-4}{48}$$

$$V = 24$$

$$\frac{2}{Rm} = \frac{2}{R} - \frac{2}{f_\ell}$$

$$\frac{2}{Rm} = -\frac{2}{-6} - \frac{2}{6} = \frac{-2}{3}$$

$$Rm = -3$$

$$\frac{1}{V} + \frac{1}{-12} = \frac{-2}{3}$$

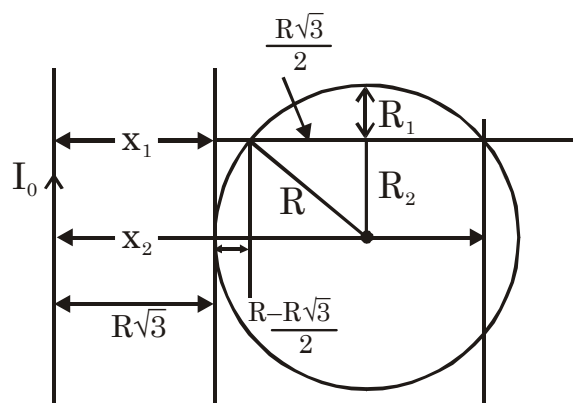
$$\frac{1}{V} + \frac{-4 \times 2}{4 \times 3} = \frac{1}{12}$$

$$V = -\frac{12}{7}$$

13. Ans. (C)

14. Ans. (D)

Sol. For (13 & 14)



$$de = \int_{x_1}^{x_2} \frac{\mu_0 i_0 V_0}{2\pi x^2} dx$$

$$2 \frac{\mu_0 i_0}{4\pi} \ln \left(\frac{2+3\sqrt{3}}{2+\sqrt{3}} \right)$$

$$x_1 = \frac{R+R\sqrt{3}}{2} = (2+\sqrt{3}) \frac{R}{2}$$

$$x_2 = R\sqrt{3} + 2R - \left(R - \frac{R\sqrt{3}}{2} \right) = \frac{(3\sqrt{3}+2)}{2} R$$

15. Ans. (B)

16. Ans. (D)

Sol. (15 and 16)

Let initially N_0 nuclei of A are present and at any time t the number of nuclei of

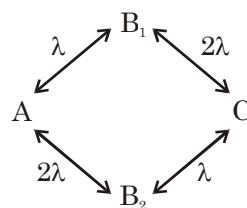
A is N_A

B_1 is N_{B_1}

B_2 is N_{B_2}

C is N_C , then

$$\frac{dN_A}{dt} = -\lambda N_A - 2\lambda N_A \Rightarrow N_A = N_0 e^{-3\lambda t}$$



$$\frac{dN_{B_1}}{dt} = \lambda N_A - 2\lambda N_{B_1} \text{ and}$$

$$\frac{dN_{B_2}}{dt} = 2\lambda N_A - \lambda N_{B_2}$$

when N_{B_1} reaches maximum value,

$$\frac{dN_{B_1}}{dt} = 0$$

$$\frac{N_A}{N_{B_1}} = 2$$

when N_{B_2} reaches maximum value,

$$\frac{dN_{B_2}}{dt} = 0$$

$$\Rightarrow \frac{N_A}{N_{B_2}} = \frac{1}{2}$$

SECTION-II

1. Ans. 5.00

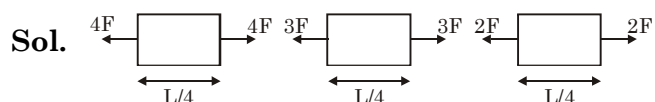
Sol. $\frac{\sqrt{4m \times 2 \times 2.2}}{2qB} = \frac{\sqrt{2mk \times 2}}{q \times 2.13B}$

$$\sqrt{k} = \frac{\sqrt{2} \times \sqrt{2.2} \times 2.13}{2\sqrt{2}}$$

$$k = 2.2 \times 4.53$$

$$k = 4.96$$

2. Ans. 3.00

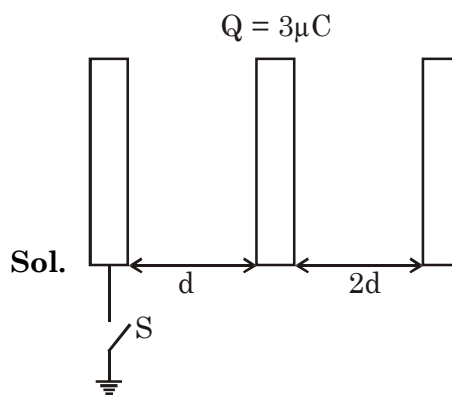


$$\frac{4F}{Ay} \times \frac{L}{4} + \frac{3F}{AY} \times \frac{L}{2} + \frac{2F}{YA} \times \frac{L}{4}$$

$$\frac{FY}{AY} \left[\frac{2+3}{2} + \frac{1}{2} \right] = 3 \frac{FL}{YA}$$

3. Ans. 11.00

4. Ans. 3.00

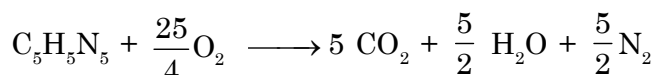


PART-2 : CHEMISTRY**SOLUTION****SECTION-I**1. **Ans.(B, C, D)**

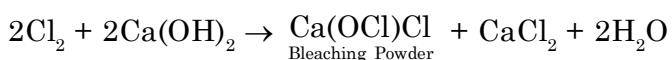
$$[H^+] = \frac{400 \times 0.1 + 500 \times 0.3 \times 2}{1000} = 0.34 \text{ M}$$

$$[Cl^-] = \frac{400 \times 0.1}{1000} = 0.04 \text{ M}$$

$$[K^+] = [NO_3^-] = \frac{100 \times 0.1}{1000} = 0.01 \text{ M}$$

2. **Ans.(C)**

$$\text{n.f of } C_5H_5N_5 = \frac{25}{4} \times 4 = 25$$

3. **Ans.(A, C, D)**4. **Ans.(B,D)**5. **Ans.(B,C)**6. **Ans.(A,B,C)**7. **Ans.(A,B,C)****Sol.**Ca²⁺ gives brick red colour in flame8. **Ans.(A,B,D)****Sol.** Ionisation energy of N→P→As is higher than group 16 elements of corresponding period9. **Ans.(B,C)****Sol.** In (B) and (C) option MnO₄⁻ act as oxidising agent.10. **Ans.(A,B,C,D)****Sol.** When alkali metal dissolves in liquid ammonia then ammoniated metal ion and ammoniated electron are formed, their dilute solution is dark blue and metal can be recovered by vaporization of NH₃.11. **Ans.(D)**

$$\text{gm equivalents} = \frac{I \times t}{96500} = \frac{96.5 \times 100}{96500} \times 0.8 = 0.08$$

$$n_{H_2} \times 2 = 0.08$$

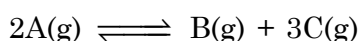
$$n_{O_2} \times 4 = 0.08$$

12. **Ans.(D)**

$$\frac{T_2}{T_1} = \left(\frac{V_1}{V_2} \right)^{\gamma-1} = \left(\frac{1}{32} \right)^{2/5} = \frac{1}{4}$$

$$T_2 = \frac{300}{4} = 75 \text{ K}$$

$$W = nC_v = (T_2 - T_1) = 0.06 \times \frac{5}{2} R(300 - 75)$$

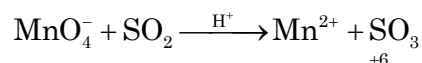
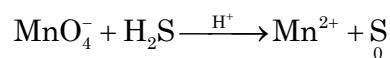
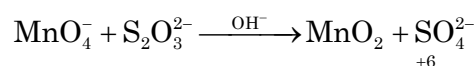
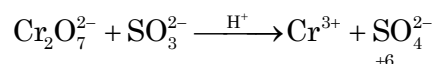
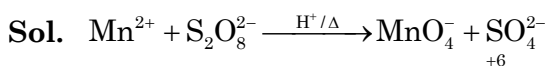
13. **Ans.(A)**14. **Ans.(B)**15. **Ans.(B)****Sol.** Zinc is extracted by roasting of sphalerite (ZnS) followed by carbon reduction.16. **Ans.(D)****Sol.** (D) Copper matte ; Cu₂S & Cu₂O ; Blister copper**SECTION-II**1. **Ans.(3.00)**

$$1 \quad \quad \quad - \quad \quad \quad -$$

$$1 - \alpha \quad \quad \quad \frac{\alpha}{2} \quad \quad \quad \frac{3\alpha}{2}$$

$$\bar{M} = 50 = \frac{70}{1 + \alpha}$$

$$\alpha = \frac{2}{5}$$

2. **Ans.(4.00)**3. **Ans.(3.00)**4. **Ans.(+24.00)**

$$6 + 6 + 6 + 0 + 6 = 24.00$$

PART-3 : MATHEMATICS

SOLUTION

SECTION-I

1. **Ans. (A,B,D)**

Sol. $\therefore f(2x) = f(x)$

put $x \rightarrow \frac{x}{2}$

$$\therefore f(2x) = f(x) = f\left(\frac{x}{2}\right) = f\left(\frac{x}{2^2}\right) = \dots = f\left(\frac{x}{2^n}\right)$$

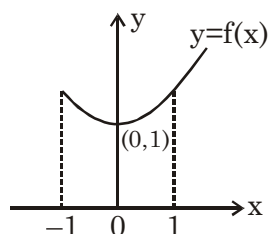
$$\therefore f(x) = \lim_{n \rightarrow \infty} f\left(\frac{x}{2^n}\right) = f(0) = \text{constant } \forall x \in \mathbb{R}$$

$$\therefore f(2) = 1$$

$\Rightarrow f(x) = 1$ i.e. constant function

2. **Ans. (A,B,C,D)**

Sol. $f(x) = \int \frac{(x+1)-1}{(x+1)^2} \cdot e^x dx = \frac{e^x}{x+1} + \lambda$



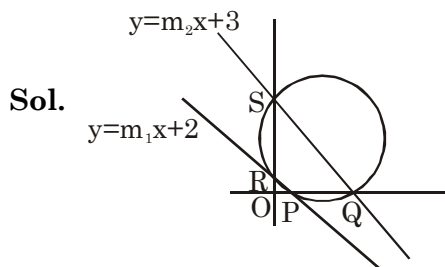
$$\therefore f(0) = 1 \Rightarrow \lambda = 0 \Rightarrow f(x) = \frac{e^x}{x+1}$$

$$\& f'(x) = \frac{x e^x}{(x+1)^2}$$

$\therefore f(x)$ is monotonic decreasing in $(-1,0)$

$$\therefore f\left(-\frac{1}{2}\right) > f\left(-\frac{1}{3}\right)$$

3. **Ans. (A,B,D)**



$$OP \cdot OQ = OR \cdot OS$$

$$\Rightarrow m_1 m_2 = 1$$

equation of concyclic circle is

$$(m_1 x - y + c_1)(m_2 x - y + c_2) + \lambda xy = 0$$

(Here $c_1 = 2, c_2 = 3$)

$$m_1 m_2 x^2 + y^2 + xy(\lambda - m_1 - m_2)$$

$$+ x(m_1 c_2 + m_2 c_1) - y(c_1 + c_2) + c_1 c_2 = 0$$

above is circle if $m_1 m_2 = 1$ & $\lambda = m_1 + m_2$

$\therefore$ equation of circle is

$$x^2 + y^2 + x(3m_1 + 2m_2) - 5y + 6 = 0$$

$$\text{length of intercept on x-axis} = 2\sqrt{g^2 - c}$$

$$= \sqrt{(3m_1 + 2m_2)^2 - 24}$$

$$\text{If } m_1 = 2 \Rightarrow m_2 = \frac{1}{2} \Rightarrow \text{length} = 5$$

4. **Ans. (B)**

Sol. $\frac{2}{b} = \frac{1}{a} + \frac{1}{c}$ where $a = 10, c = kb, k \in \text{integer}$

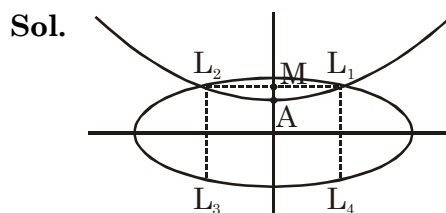
$$\therefore k = \frac{10}{20-b}$$

$$\text{when } b = 15, \quad k = 2$$

$$b = 18, \quad k = 5$$

$$b = 19, \quad k = 10$$

5. **Ans. (A,B,D)**



$$L_1\left(3, \frac{16}{5}\right), L_2\left(-3, \frac{16}{5}\right) \text{ are extremity of parabola whose focus is M}$$

$\therefore$ length of latus rectum

$$= 4a = 6 \Rightarrow a = \frac{3}{2} = MA$$

$$\therefore M\left(0, \frac{16}{5}\right)$$

$$\therefore \text{vertex } A\left(0, \frac{16}{5} - \frac{3}{2}\right) \equiv A\left(0, \frac{17}{10}\right)$$

$$\therefore \text{vertex } A\left(0, \frac{16}{5} - \frac{3}{2}\right) \equiv A\left(0, \frac{17}{10}\right)$$

$$\text{foot of directrix } N\left(0, \frac{16}{5} - 3\right) \equiv N\left(0, \frac{1}{5}\right)$$

$$\therefore \text{equation of directrix is } y = \frac{1}{5}$$

equation of hyperbola confocal to ellipse is

$$\frac{x^2}{a^2 - k^2} - \frac{y^2}{k^2 - b^2} = 1$$

so take $k^2 = 20$ (Here $a^2 = 25, b^2 = 16$)

$$\therefore \frac{x^2}{5} - \frac{y^2}{4} = 1$$

6. Ans. (A,C,D)

Sol. option 1, $P(E) = \frac{|6|5|}{|10|}$

option 2, $P(E) = \frac{|2|5|5|}{|10|}$

option 3, $P(E) = \frac{|2|5|5|}{|6|5|} = \frac{1}{3}$

option 4, $P(E) = \frac{|2|4|4|9|}{2|5|5|} = \frac{9}{25}$

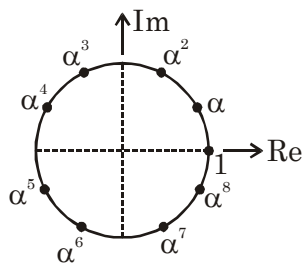
7. Ans. (A,B,D)

Sol. $z^9 - 1 = (z - 1)(z^8 + z^7 + \dots + z + 1)$
 $= (z - 1)(z - \alpha)(z - \alpha^2) \dots (z - \alpha^8)$
 (Because roots are in G.P.)

$\therefore (z - \alpha)(z - \alpha^2) \dots (z - \alpha^8) = \frac{z^9 - 1}{z - 1}$

Put $z = 2 \Rightarrow (2 - \alpha)(2 - \alpha^2) \dots (2 - \alpha^8) = 511$

$\sum_{\alpha=1}^4 \operatorname{Re}(\alpha^i) = \cos 40^\circ + \cos 80^\circ + \cos 120^\circ + \cos 160^\circ$
 $= -\frac{1}{2}$

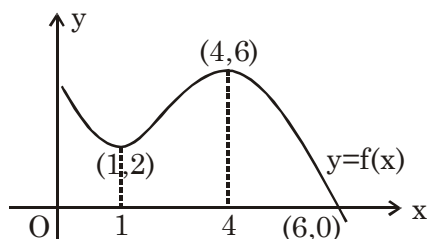


option 3 $\frac{\sum_{i=0}^8 |\alpha^i - \alpha^{i+1}|}{\sum_{i=0}^4 |\alpha^{2i} - \alpha^{2i+1}|} = \frac{9\lambda}{5\lambda} = \frac{9}{5}$

option 4 $\sum_{i=1}^4 \alpha^i \alpha^{9-i} = \alpha^1 \alpha^8 + \alpha^2 \alpha^7 + \alpha^3 \alpha^6 + \alpha^4 \alpha^5 = 4$

8. Ans. (A,B,C,D)

Sol.



Curve will not intersect x-axis
 one local max & one local min
 $y = f(x)$ is increasing in $x \in (1, 4)$

$\therefore \int_1^4 f(x) dx \in \text{min area \& max area}$

$\therefore (4-1) \cdot 2 < \int_1^4 f(x) dx < (4-1) \cdot 6$

$f(1^+) = 2^+$ & $f(4^-) = 6^-$

Now check option 4

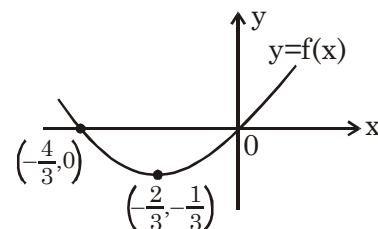
9. Ans. (A,C,D)

Sol. $f(x) = x + x^2 \int_0^1 f(t) dt$

$f(x) = x + \lambda x^2$ where $\lambda = \int_0^1 f(t) dt$

$\lambda = \int_0^1 (x + \lambda x^2) dx$

$\Rightarrow \lambda = \frac{3}{4}$



$\therefore f(x) = \frac{3}{4}x^2 + x$

Area bounded $= \int_{-4/3}^0 \left(\frac{3}{4}x^2 + x \right) dx = -\frac{8}{27}$

$f\left(\frac{1}{x^2}\right) = \frac{3}{4x^4} + \frac{1}{x^2}$

$\therefore \lim_{x \rightarrow \infty} f(x) \cdot f\left(\frac{1}{x^2}\right) = \lim_{x \rightarrow \infty} \frac{\left(\frac{3}{4}x^2 + x\right)(3 + 4x^2)}{4x^4} = \frac{3}{4}$

10. Ans. (A,B,C,D)

Sol. For C_1 $CC_1 = r + 1$, $CC_2 = r + 5$
 $\Rightarrow CC_2 - CC_1 = 4$

$$2a = 4 \text{ \& } C_1C_2 = 2ae$$

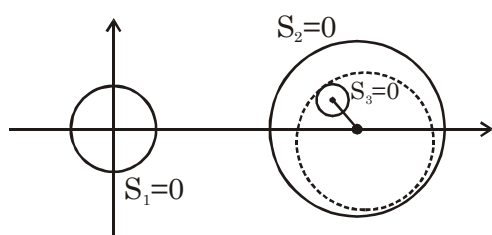
$$\Rightarrow e = 2$$

For C_2 $CC_3 = r + 1$, $CC_2 = 5 - r$
 $CC_2 + CC_3 = 6$

$$\therefore 2a = 6 \text{ \& } C_2C_3 = 2ae$$

$$\Rightarrow a = 3 \text{ \& } e = \frac{1}{3\sqrt{2}}$$

For C_3 is parabola $\therefore e = 1$



If S is largest circle as shown in the figure

$$\therefore \text{its diameter} = 1 + \sqrt{2} + 5$$

$$= 6 + \sqrt{2}$$

Paragraph for Question 11 and 12

11. Ans. (C)

12. Ans. (D)

Sol. Let $M(-1 + \lambda, -2 + \lambda, -3 + \lambda)$ on line L_2 such that $OM \perp$ line L_1

$$\therefore (-1 + \lambda).1 + (-2 + \lambda).1 + (-3 + \lambda).1 = 0$$

$$\Rightarrow \lambda = 2$$

$$\therefore M(1, 0, -1)$$

Let line L_1 & L_3 lie on plane P_1 , then normal to plane P_1 is $(i - k)$ & equation of plane is $x - z = 0$

$$\therefore \text{dr of line ON} = \begin{vmatrix} i & j & k \\ 1 & 0 & -1 \\ 1 & 1 & 1 \end{vmatrix} = i - 2j + k$$

mid point of P & M is origin $\Rightarrow P$ is $(-1, 0, 1)$

Paragraph for Question 13 and 14

13. Ans. (D)

Sol. diagonal elements must be $\langle 0, 0, 0 \rangle$

$$\begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & -1 \\ -1 & 1 & 0 \end{bmatrix}$$

Now above diagonal element we can place

case 1 : $0, 1, -1$ by $\frac{3}{2}$ ways

case 2 : $0, 1, 1$ by $\frac{3}{2}$ ways

case 3 : $0, -1, -1$ by $\frac{3}{2}$ ways

$\therefore$ Total number of skew symmetric matrix = 12

14. Ans. (C)

Sol. $\Delta = \begin{bmatrix} k_1 & a & b \\ a & k_2 & c \\ b & c & k_3 \end{bmatrix}$ be the corresponding

matrix from set A. Its determinant value

$$\Delta = k_1k_2k_3 - (k_1c^2 + k_2b^2 + k_3a^2) + 2abc$$

case 1 : all diagonal element must be 0

$$\Rightarrow \Delta = 0$$

so all matrix is singular

case 2 : Only one diagonal element is 0 say

$$k_1 = 0$$

$$\Rightarrow \Delta = 2abc - k_2b^2 - k_3a^2$$

$$\begin{array}{|c|c|c|} \hline a = b = 0 & a = c = 0 & b = c = 0 \\ \hline \end{array}$$

$$\Delta = 0 \quad \Delta = k_2b^2 \quad \Delta = -a^2k_3$$

here two

here two

non-singular

non-singular

matrix possible

matrix possible

$\therefore$ Total 4 matrix when $k_1 = 0$

4 matrix when $k_2 = 0$ and $k_3 = 0 \therefore$ Total = 12

Paragraph for Question 15 and 16

15. Ans. (C)

16. Ans. (C)

Sol. $(x+1) \int_0^x f(t) dt - \int_0^x t f(t) dt = x^4 + x^2$

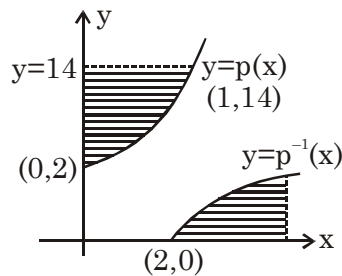
differentiate wrt x,

$$f(x) + \int_0^x f(t) dt = 4x^3 + 2x$$

differentiate wrt x, $f'(x) + f(x) = 12x^2 + 2$

I.F = e^x

$$\therefore f(x) = 12x^2 - 24x + 26 + \lambda e^{-x}$$



$$\therefore f(0) = 26 \Rightarrow \lambda = 0$$

$$\Rightarrow f(x) = 12x^2 - 24x + 26$$

$$f'(x) = 0 \Rightarrow x = 1$$

$\therefore f_{\min}$ at $x = 1$ and its value is 14

$$\text{Area bounded} = \int_0^1 (14 - 12x^2 - 2) dx$$

$$= 12 \int_0^1 (1 - x^2) dx = 8$$

SECTION-II

1. Ans. 0.20

Sol. ${}^m C_0 {}^m C_3 + {}^m C_1 {}^m C_4 + {}^m C_2 {}^m C_5 + \dots + {}^m C_{m-3} {}^m C_m$
 ${}^m C_m = 2^m C_{m-3} = 45$
 then $m = 5$

2. Ans. 1.50

Sol. $x^3 + 3x^2 + 5x + 3 = 0$ has one root $x = -1$

$$\therefore x^3 + 3x^2 + 5x + 3 = (x+1)(x^2 + 2x + 3)$$

$$\Rightarrow a = 2, b = 3$$

3. Ans. 1.33 or 1.34

Sol. $\vec{u} \times \vec{v} = \vec{w} \Rightarrow \vec{u} \times \vec{v} \cdot \vec{w} = \vec{w} \cdot \vec{w}$

$$\Rightarrow [\vec{u} \vec{v} \vec{w}] = 1 \quad \dots (i)$$

$$\text{take } \vec{a} = x\vec{u} + y\vec{v} + z\vec{w}$$

take dot product with $\vec{u} \times \vec{v}$, $\vec{v} \times \vec{w}$ & $\vec{w} \times \vec{u}$, we get

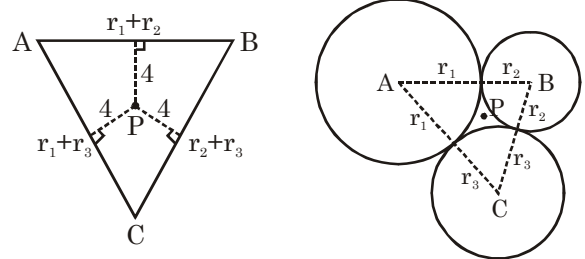
$$\Rightarrow x = 2, y = 3, z = 4$$

$$\therefore \vec{a} = 2\vec{u} + 3\vec{v} + 4\vec{w}$$

$$\therefore \frac{\text{projection of } \vec{a} \text{ upon } \vec{w}}{\text{projection of } \vec{a} \text{ upon } \vec{v}} = \frac{4}{3} = 1.33$$

4. Ans. 1.33 or 1.34

Sol.



$$\Delta PAC + \Delta PBC + \Delta PAB = \Delta ABC$$

$$\frac{1}{2} \cdot 4(r_1 + r_3) + \frac{1}{2} \cdot 4(r_2 + r_3) + \frac{1}{2} \cdot 4(r_1 + r_2)$$

$$= \sqrt{(r_1 + r_2 + r_3) r_1 r_2 r_3}$$

$$\Rightarrow 16(r_1 + r_2 + r_3) = r_1 r_2 r_3 \quad \dots (i)$$

$$\text{take } r_1 = n, r_2 = n+1, r_3 = n+2$$

Put in equation (i), we get $n = 6$

$$\therefore \text{radius of three circle} = 6, 7, 8$$

$$\therefore \frac{r_{\text{largest}}}{r_{\text{smallest}}} = \frac{8}{6} = 1.33$$